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Claude Code with Git Worktrees

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Introducing Today's Project!

In this project, I'm going to accelerate my development. This will help me build and ship products at a faster pace. I'm interested in this because I am trying to upskill my Claude Code and vibe coding skills.

Key tools and concepts

Git is a tool that allows for verison control. It creates snapshots of the code in branches, and developers can use these snapshots to develop and rollback the code whenever something breaks. It also allows for collaboration by making it possible for multiple devs to work on a codebase (each dev gets their own branch). We need it for this project because we are trying to build multiple features which are not related to one another.

Challenges and wins

A commit is saving a snapshot of your project or a record of the project. It is useful because it allows you to save your work and progress.

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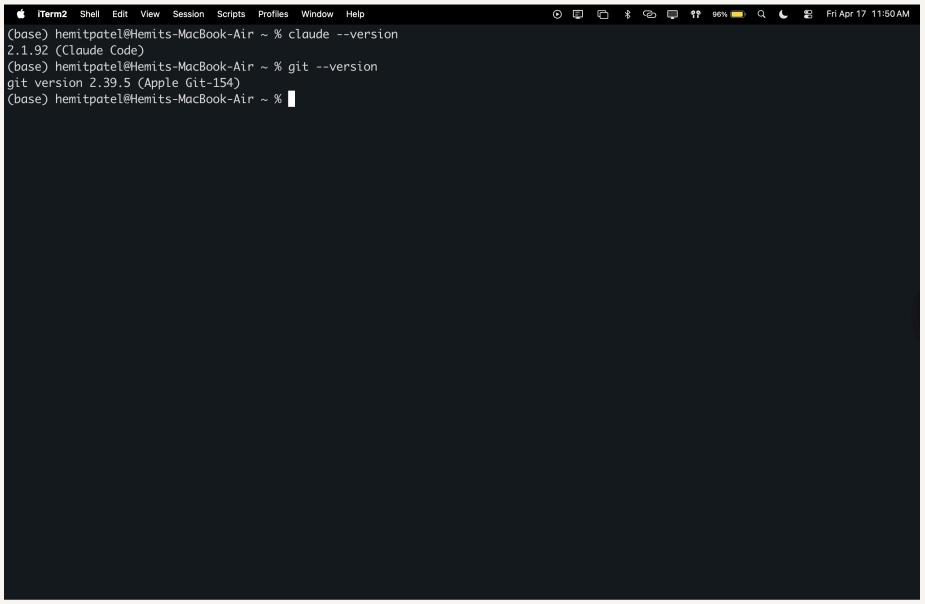
Installing Claude Code and Git

In this step, I am setting up Claude Code and Git using homebrew CLI tool.



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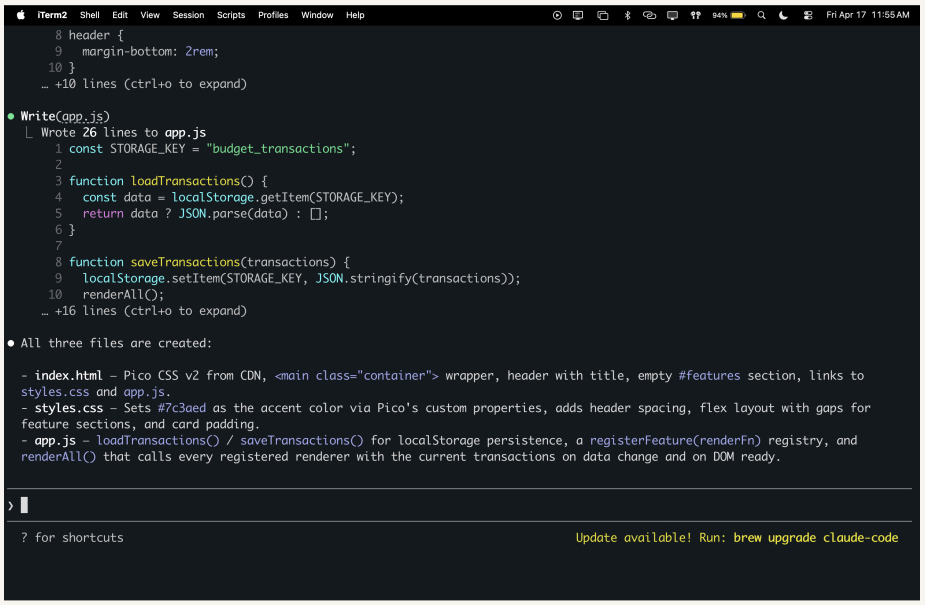
Scaffolding the Budget Tracker with AI

In this step, I am creating the project folder and initializing a Git repository and using Claude Code to scaffold the budget tracker app (so that we know what features to build and the overall structure of the software). And we need to understand Git concepts to make it easier to build out solutions.



Understanding the project structure

Claude Code created three files: index.html, styles.css, app.js.The empty main element is where we will be creating the 3 main features that require 3 seperate git worktrees.

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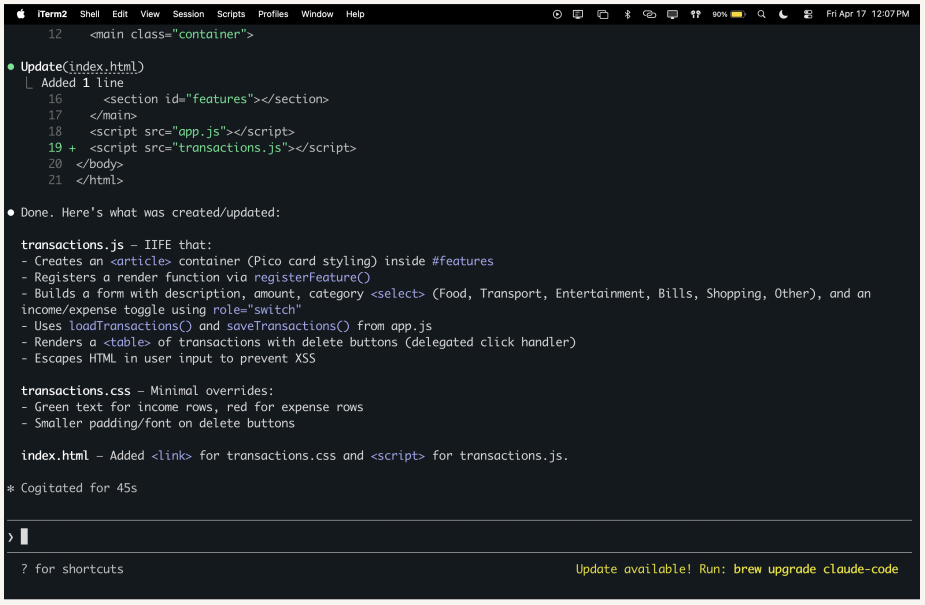
Launching the First Worktree Session

In this step, I'm going to launch a worktree session with Claude Code. We will be adding a transaction feature using these coding tools.



How worktree isolation works

I used worktrees because I want to build features in parallel. The worktree created a new branch called add-transactions and a folder at .claude/worktrees

```
(base) hemitpatel@Hemits-MacBook-Air budget-tracker % git branch
* main
+ worktree-add-transactions
```

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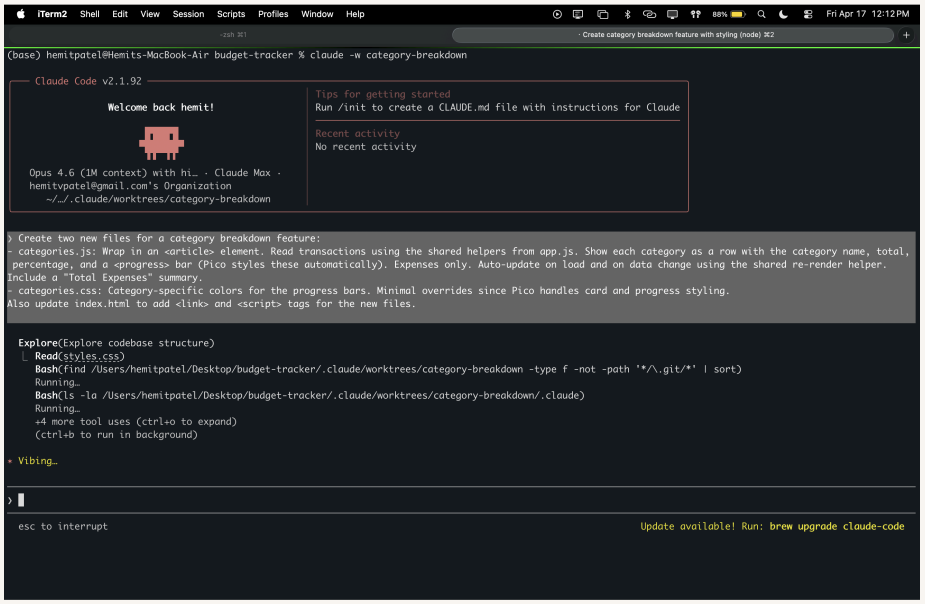
Running Parallel Claude Code Sessions

In this step, I am launching a second CC worktree session for category breakdown and a third CC worktree for the monthly summary. Both of these git worktrees will run in parallel.



Why worktrees stay independent

Just like in Step 3, this worktree creates a new branch called category-breakdown and does all changes for this branch within a new folder (not within the main folder) that lives in .claude.

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```
(base) hemitpatel@Hemits-MacBook-Air budget-tracker % git branch

* main
+ worktree-add-transactions
+ worktree-category-breakdown
+ worktree-monthly-summary
```

Managing multiple feature branches

The three feature branches I see are: + worktree-add-transactions + worktree-category-breakdown + worktree-monthly-summary

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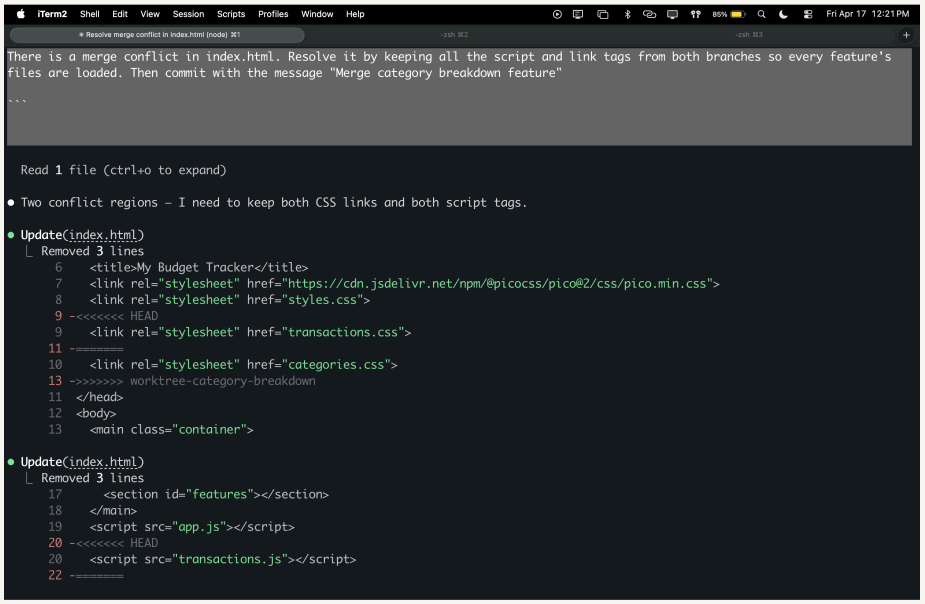
Merging Three Features into One App

I'm merging all 3 branches into main because I want to see all features working together. Then we resolve any merge conflicts that have occured with Claude Code and test the budget tracke with the browser.



Resolving merge conflicts with AI

A merge conflict happens when 2 branches write to the same line on the same file Claude resolved it by removing the bad change and understanding how to better write the code.

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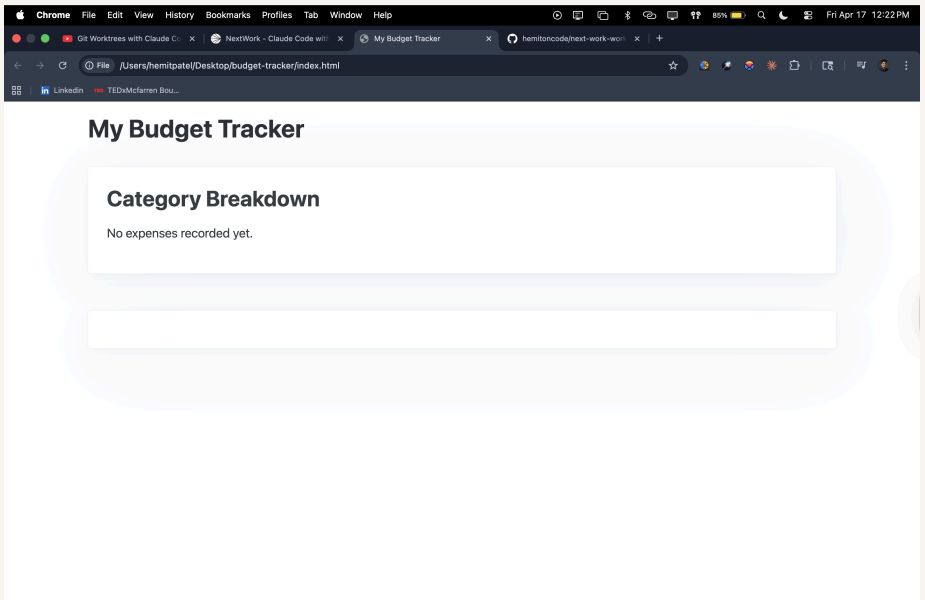

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